# Your grade: 96%

Your latest: **93.33%** • Your highest: **96%** • To pass you need at least 80%. We keep your highest score.

Next item

coach

**Almost there! Let's review your answers and then try again.**

Your work showed a strong understanding of neural network training processes and PyTorch functionalities. However, there is room for improvement in understanding data handling with PyTorch. Before retrying this assignment, we recommend focusing on this key area:

- **Datasets and Dataloaders (101)**: In Question 5, your understanding of the actions that can be performed using the PyTorch Dataloader class was incomplete. Try reviewing this item to refresh your memory on the capabilities and functionalities of the Dataloader class.

Good job on the rest of the assignment!

Retry assignment

1. What is the role of weights in a neural network during the training process? 1 / 1 point

- ( ) Weights are only used during forward passes and not during backward passes.
- (•) Weights are adjusted to minimize the loss function, helping the model make more accurate predictions.
- ( ) Weights remain constant and do not change during the training process.
- ( ) Weights determine the structure of the neural network and do not change once the network is defined.

**Correct**
Correct! During training, weights are updated to minimize the loss function and improve the model's predictions.

2. What does the backward() function in PyTorch do? 1 / 1 point

- (•) Computes the gradient of a tensor
- ( ) Saves the computational graph
- ( ) Performs matrix multiplication
- ( ) Initializes model parameters

**Correct**
Correct! The backward() function is used to compute the gradient of a tensor, which is crucial for backpropagation and training neural networks.

3. Which component in a PyTorch model is responsible for updating the model parameters based on the calculated gradients? 1 / 1 point

- ( ) Loss function
- ( ) Activation function
- (•) Optimizer
- ( ) DataLoader

**Correct**
Correct! The optimizer updates the model parameters based on the calculated gradients and the learning rate.

4. What is the primary purpose of partitioning data into batches during model training? 1 / 1 point

- ( ) To simplify the implementation of the model
- ( ) To increase the complexity of the model
- (•) To efficiently use computational resources and stabilize gradient updates
- ( ) To ensure different data distributions are used in training

**Correct**
Correct! Partitioning data into batches helps efficiently use computational resources and stabilizes gradient updates.

5. Which of the following actions can be performed using the PyTorch Dataloader class? 0.6666666666666666 / 1 point

- [ ] Automatically splitting the data into training and testing sets
- [ ] Loading data in parallel using multiprocessing
- [x] Shuffling the data

**Correct**
Correct! PyTorch Dataloader can shuffle the data to ensure that the model does not learn the order of the data.

- [ ] Sampling the data based on specified probabilities
- [ ] Normalizing the data
- [x] Collating data into batches

**Correct**
Correct! PyTorch Dataloader collates data into batches for efficient model training.

You didn't select all the correct answers